COURSE CODE

SBT-DF204

COURSE TITLE

Computer Forensics Case Studies

LAB TITLE

Investigating DJI Mavic Air Flight Evidence

NAME

Kumpon Emmanuel Nandom

STUDENT ID

2025/FWSD/11325

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PLATFORM USED

Kali Linux

INSTRUCTOR

Mr. Aminu Idris, AMCPN

Commandant

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Computer Forensics Case Studies

Case Study 5: Investigating DJI Mavic Air Flight Evidence

Individual report

Full name	Kumpon Emmanuel
Registration number	2025/FWSD/11325
Course code	SBT- DF204 Computer Forensics Case Studies
Case title	Investigating DJI Mavic Air Flight Evidence
Examination date	2 November 2026
Workstation	Kali Linux

Executive Summary

The Android logical image contains records from a DJI GO 4 flight on June 19, 2018. It includes a main flight record (TXT), a cached video (MP4), information files, two detailed records (DAT: FLY057 and FLY058), and six cached thumbnail images. The TXT and MP4 files were last updated 2 seconds apart (11:55:40 and 11:55:42 according to the UTC on my workstation). FLY057 was updated 2 minutes and 44 seconds after the TXT (11:58:24). This order matches the record types, the TXT and video end when the flight lands, and the detailed record ends when the device powers off.

There is a moderate level of confidence that the TXT, MP4, and FLY057 come from the same flight session. However, the accuracy of the UTC time is low. The recorded times can shift by five hours based on the time zone used when unpacking the archive. Additionally, the phone clock that labels the files is two hours off from the actual file times. No information I found identifies an operator, owner, or flight route, and I do not make any claims about these details.

Evidence acquisition and integrity

I store my work in the folder ~/SBTDF204_CS5, which has three subfolders. The "originals" folder contains the supplied files and is set to mode 444. I never use these files for analysis. The work folder holds my working copy and the extracted data. The evidence_out folder contains saved hashes, tool versions, and command output.

Source	Original file name	SHA-256	Source location	Working copy
Android logical image	Android_Logical.zip	9a352b282d832556eb7cb73010ea281f9946d889438ce07d9bbee3b64229285d	E-Campus package, copied to ~/SBTDF204_CS5/originals/	~/SBTDF204_CS5/work/Android_Logical.zip, extracted to work/Android_Logical/

I created a hash for the file Android_Logical.zip that I placed in the originals folder. I verified the hash using sha256sum -c, and it was correct. Then, I made the file read-only (S03). After that, I copied it to the work folder and hashed the copy. The hashes for both files are the same (S03, S05). I did not record a hash for the file at its original download location before copying it.

All extraction took place in the workplace. I did not upload any evidence to any website, cloud account, or scanner. No phone or drone was connected. My first stat run (S08) captured the original access times. The sha256sum run that followed read each file and changed its access time on the working copy. This is why the later stat outputs (S10, S11, S15) show an access time of 2026-11-02 21:17:33 UTC. Unpacking the archive also gave every file a creation and change time of 2026-11-02 20:24 UTC (S08).

Method

Mobile artefacts

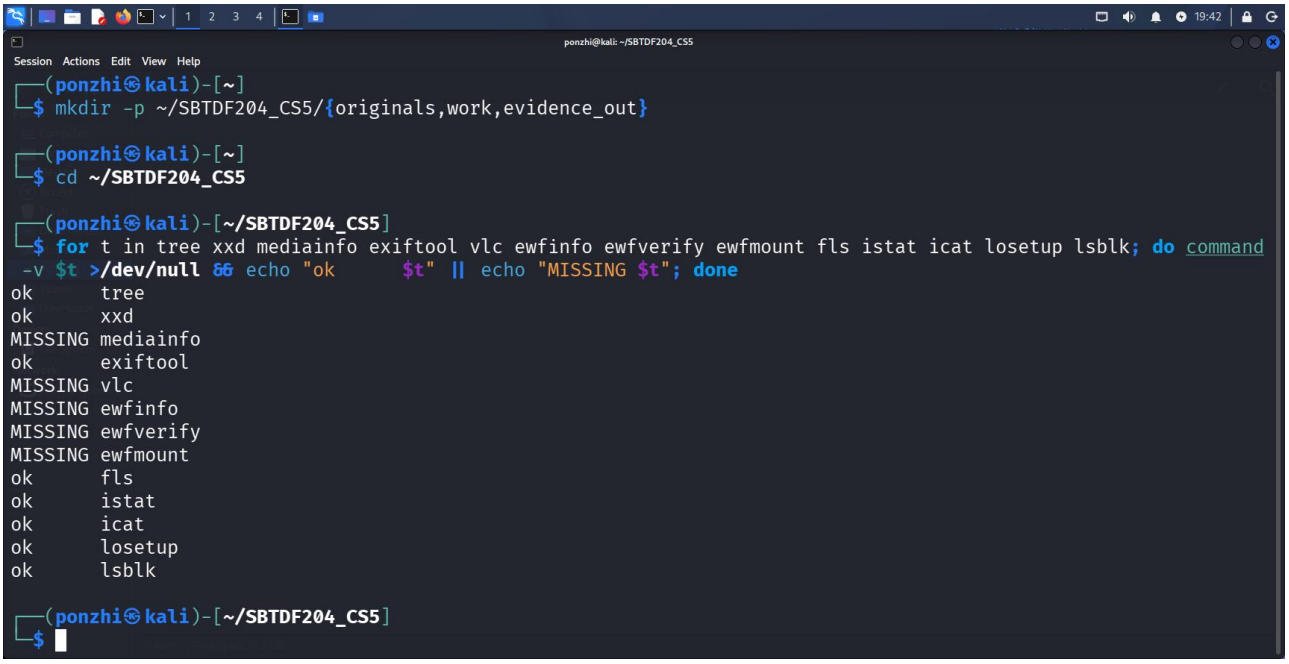
The paths below begin with Android_Logical/sdcard/DJI/dji.go.v4/, unless stated otherwise. Before opening any file, I check the file times, create a hash, and then look at the content.

Step	Command	Purpose
1	sha256sum Android_Logical.zip, sha256sum -c ../evidence_out/sha256_before.txt, chmod 444 * (in originals)	Hash the original and lock it
2	cp ../originals/* ., sha256sum *, unzip Android_Logical.zip (in work)	Working copy, integrity check, extraction
3	tree -L 1, tree FlightRecord, tree -L 2 --inodes DJI_RECORD, tree CACHE_IMAGE	Locate the four artefact families
4	TZ=UTC stat on the TXT, both DAT files, the MP4 and the six JPEGs in one command	Sizes, inodes and times before any file is opened
5	sha256sum on the same ten files	Integrity values for each artefact
6	file and xxd -s -200 on the TXT	Format and readable device text
7	mediainfo and exiftool on the MP4, cat on the .info file	Video metadata and recorded GPS
8	cp of the MP4 to preview.mp4, then vlc preview.mp4	Controlled preview of a copy
9	ls --full-time -tr, file, ls | sort -t_ -k3 -n, date -u -d @1523610990 on CACHE_IMAGE	Types, times and name-based order of the pictures

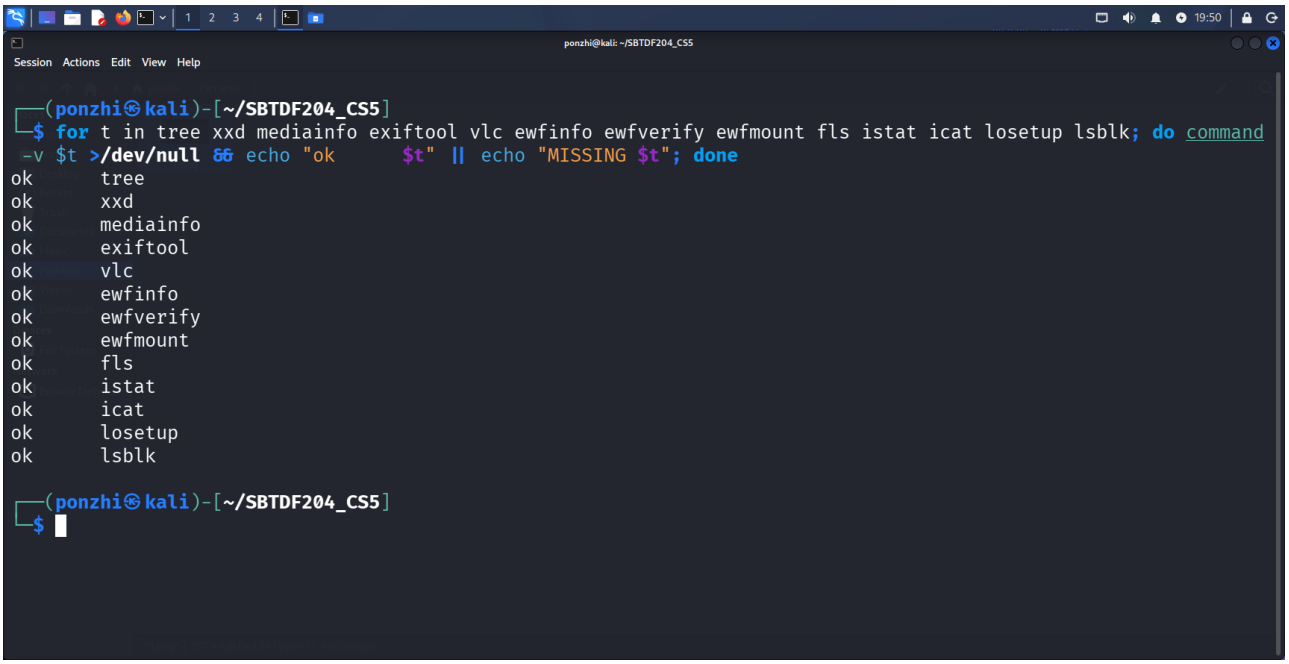
Findings

Question 1: How did you preserve the case materials?

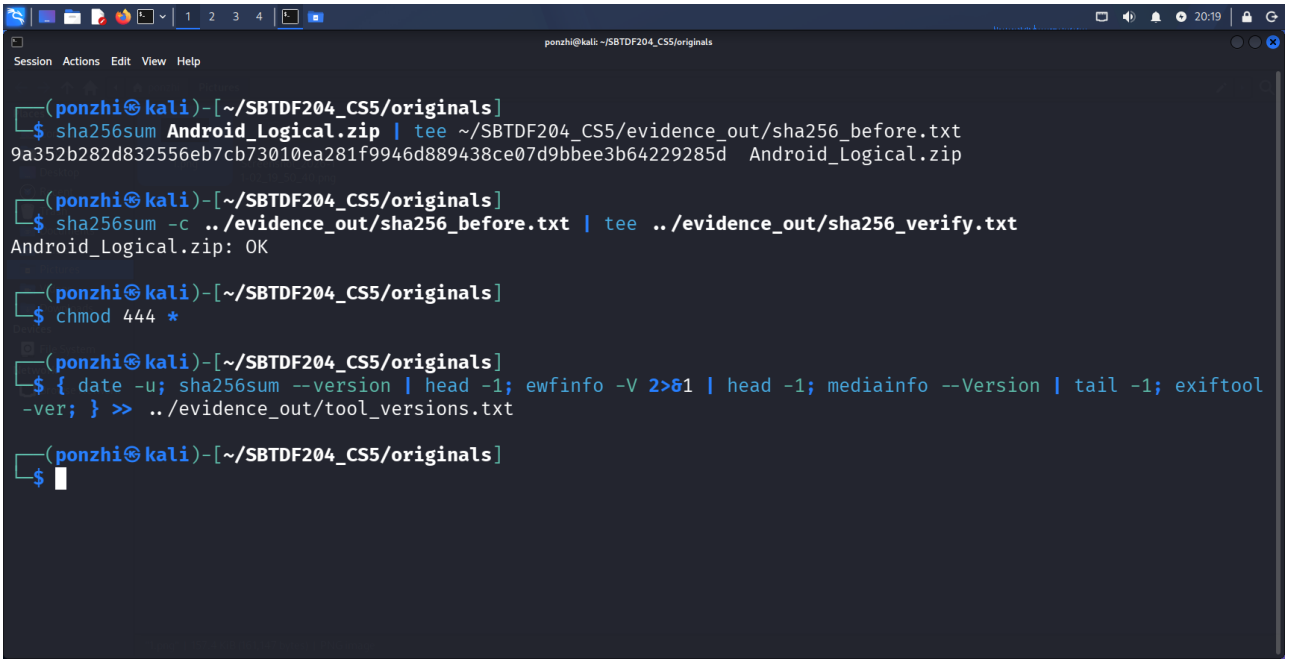
During the first run, the tools mediainfo, vlc, ewfinfo, ewfverify, and ewfmount were missing. By the second run, all thirteen tools were available. S03 shows the SHA-256 of the original file, the result of the sha256sum -c check, and changes the file permissions to read-only (chmod 444). S04 shows the saved version notes, the date was Mon 02 Nov 2026 19:19:23 UTC, the sha256sum version was GNU coreutils 9.10, ewfinfo was 20140816, and MediaInfoLib was v26.05. ExifTool version 13.55 is in S12, and VLC version 3.0.23 appears in S14. S05 shows the copy made for work, its matching hash, and the extraction process.



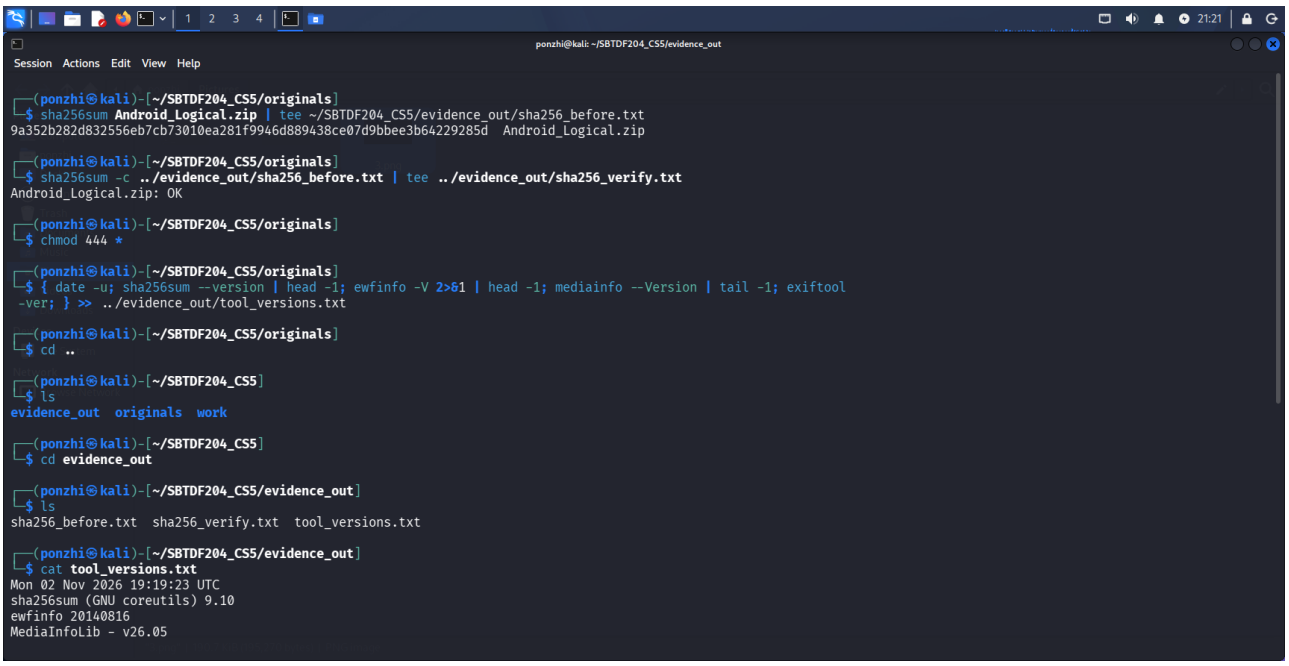
S01. Tool availability check, first run. Five tools reported MISSING.



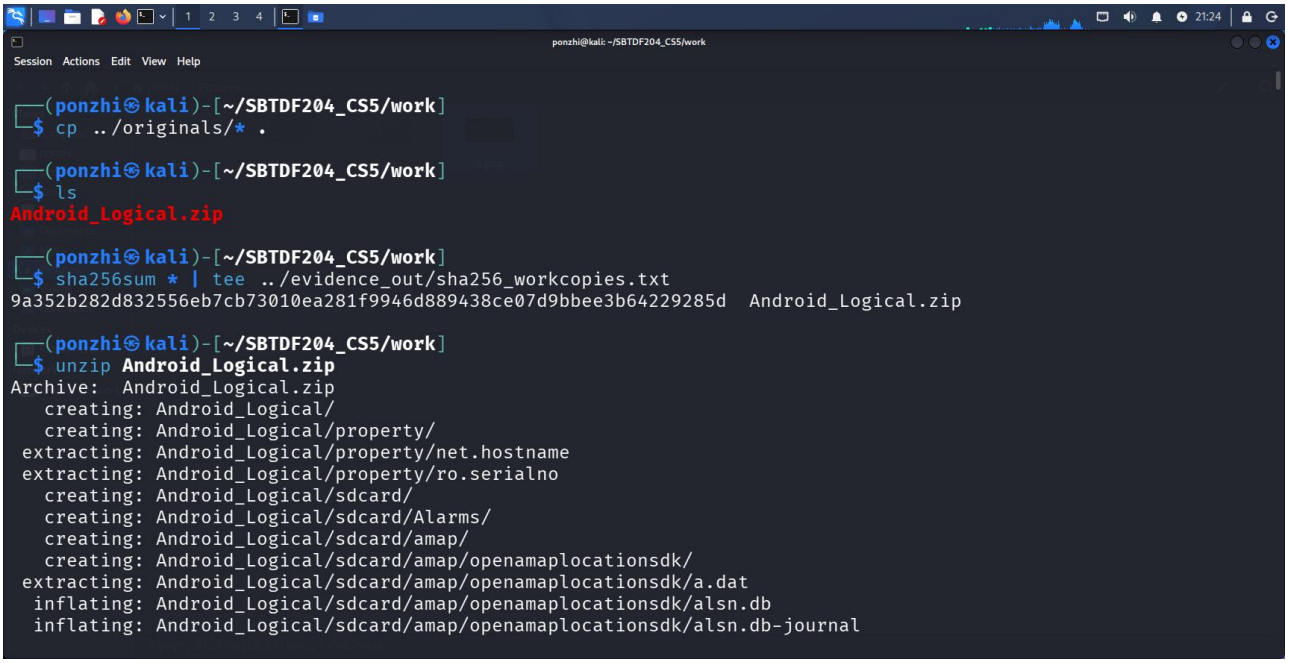
S02. Tool availability check, second run. All thirteen tools report ok.



S03. SHA-256 of Android_Logical.zip in originals, sha256sum -c result OK, chmod 444, version command.



S04. Saved tool-version notes: UTC date, sha256sum 9.10, ewfinfo 20140816, MediaInfoLib v26.05.



S05. Working copy hash matches the original, unzipped Android_Logical.zip.

Question 2: What DJI GO 4 artefacts are present?

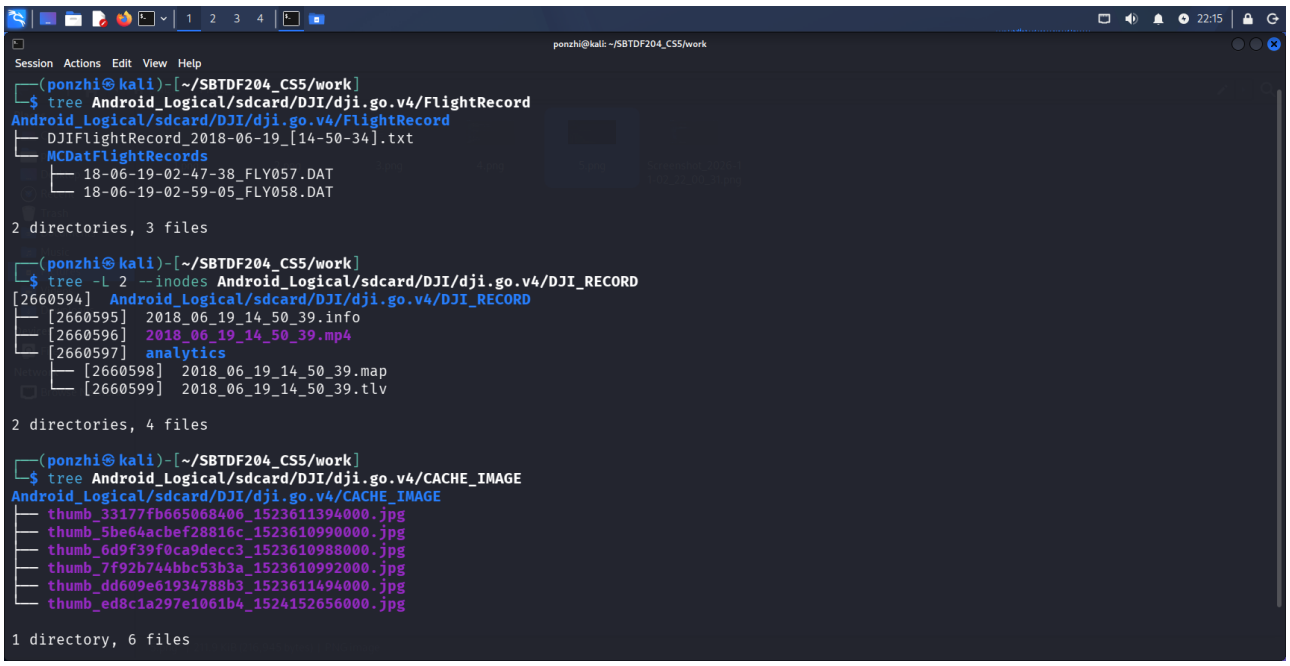
The app data is located in Android_Logical/sdcard/DJI/dji.go.v4/. It has 13 subfolders, but tree shows 14 directories because it counts the main folder. Four of these subfolders contain the artifacts described in the course.

Artefact family	Path (after the app folder)	Files found	Course scope
Main flight record	FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	1 TXT	Motors on to landed
Cached video	DJI_RECORD/2018_06_19_14_50_39.mp4	1 MP4, plus .info, and analytics/ with .map and .tlv	Take-off to landed
Detailed flight records	FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT	2 DAT	Power on to power off
Cached pictures	CACHE_IMAGE/thumb_*.jpg	6 JPEG	Not a flight record

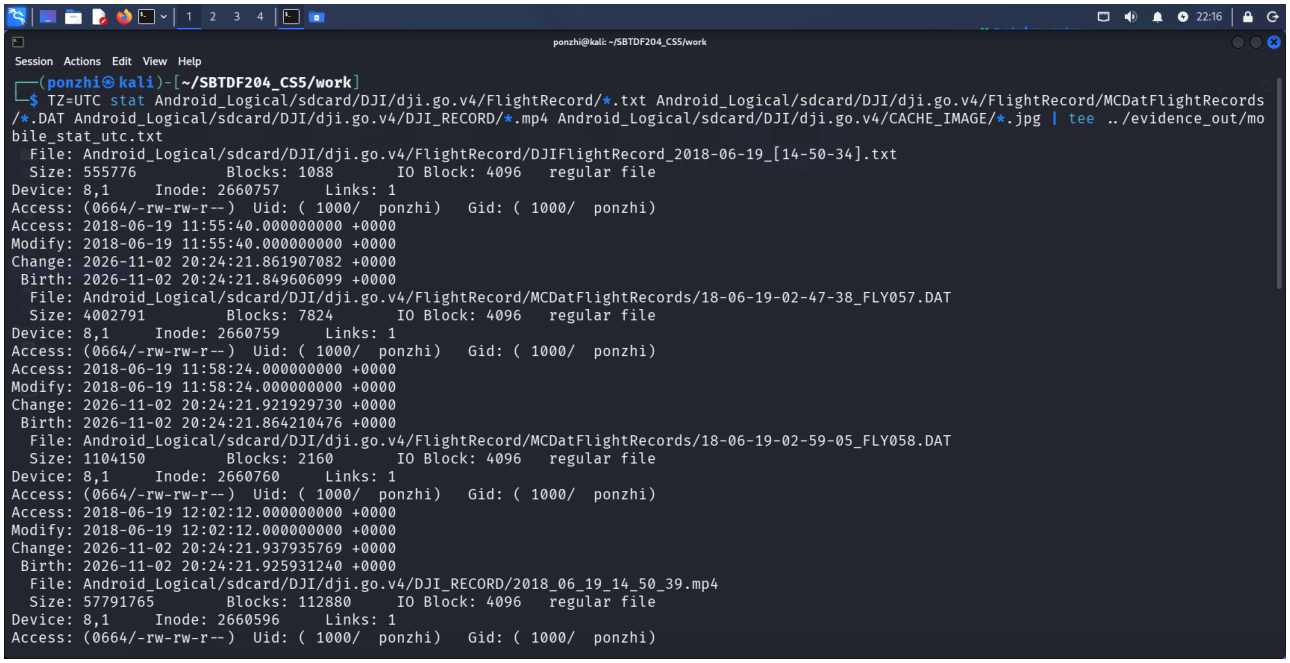
The sizes, inodes, last-write times, and SHA-256 values for all ten files are listed in Appendix B (S08, S08a, S08b, S09). The extraction also includes a folder with the net: hostname and ro.serialno (S05). I did not open these files.



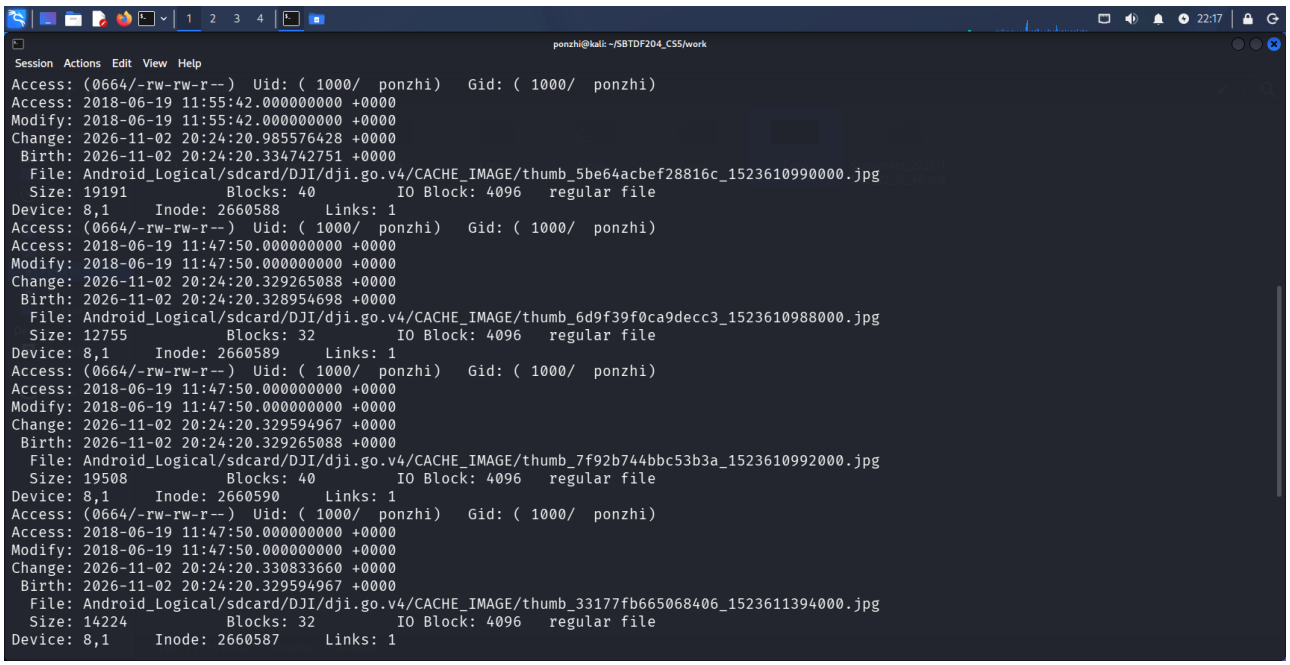
S06. tree -L 1 of the dji.go.v4 folder: 13 subfolders.



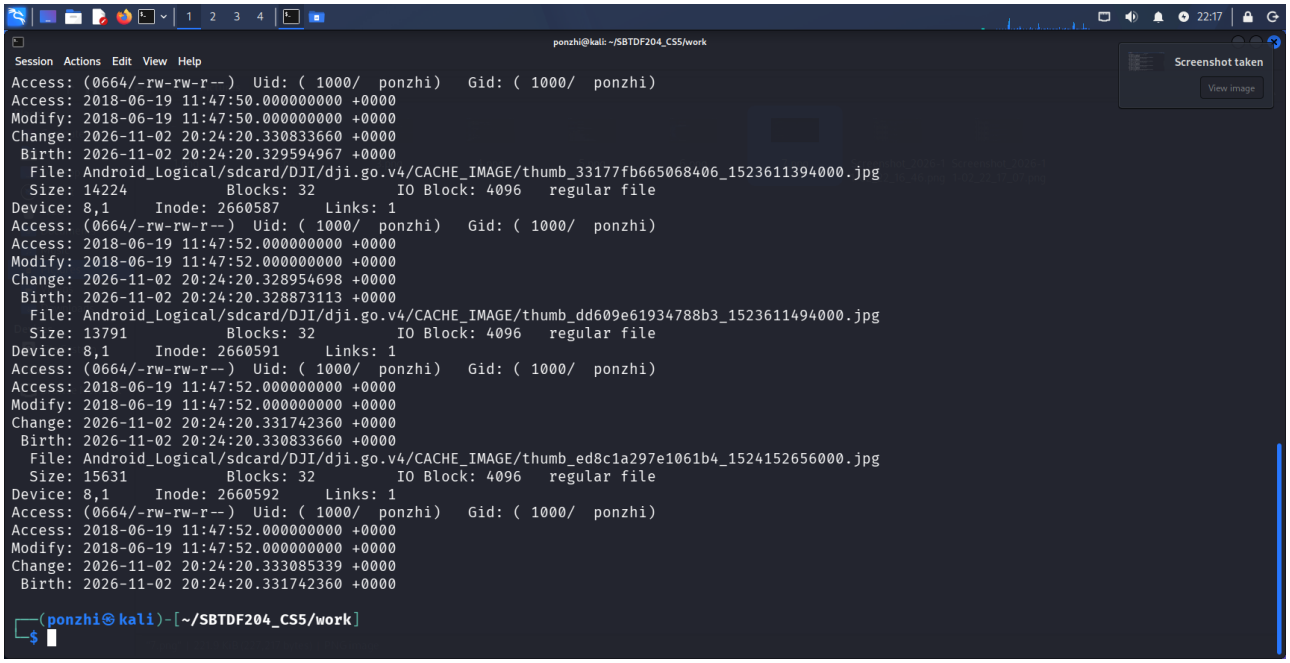
S07. tree of FlightRecord, DJI_RECORD, and CACHE_IMAGE



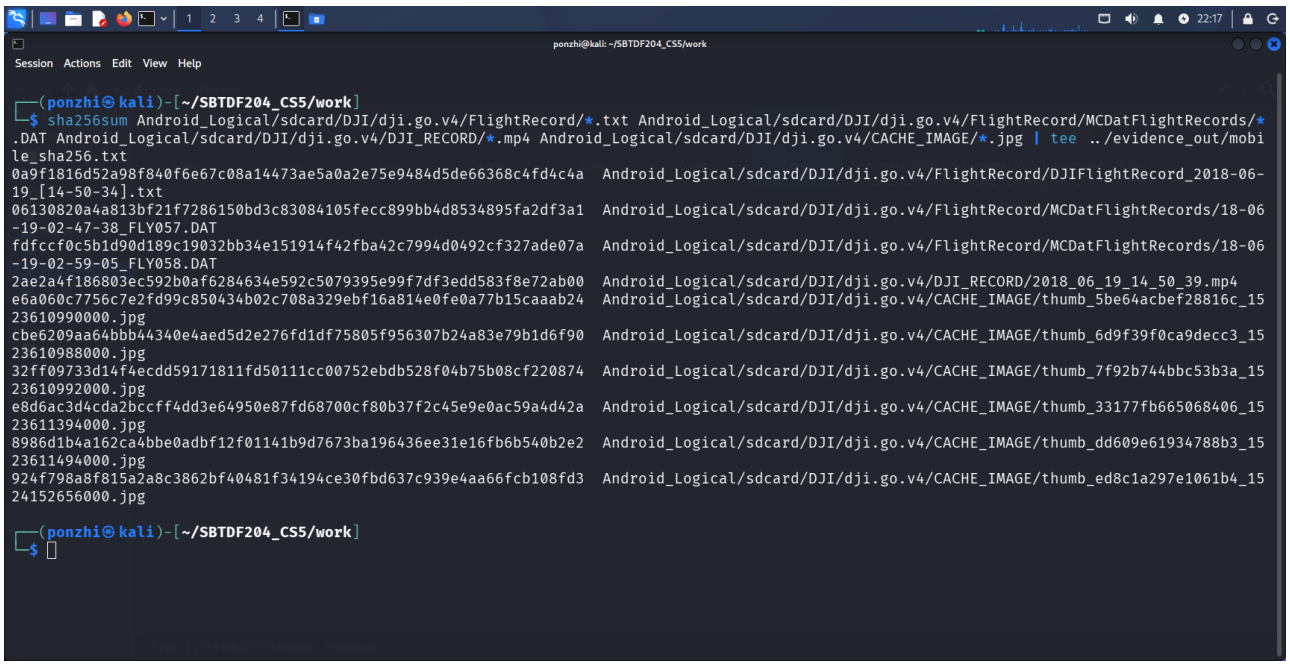
S08. TZ=UTC stat on all artefacts, part 1: TXT, FLY057, FLY058 and the start of the MP4.



S08a. TZ=UTC stat, part 2: end of the MP4 and the first three thumbnails.



S08b. TZ=UTC stat, part 3: remaining thumbnails.



S09. sha256sum of the TXT, both DAT files, the MP4 and the six JPEGs.

Question 3: What does the main flight record establish?

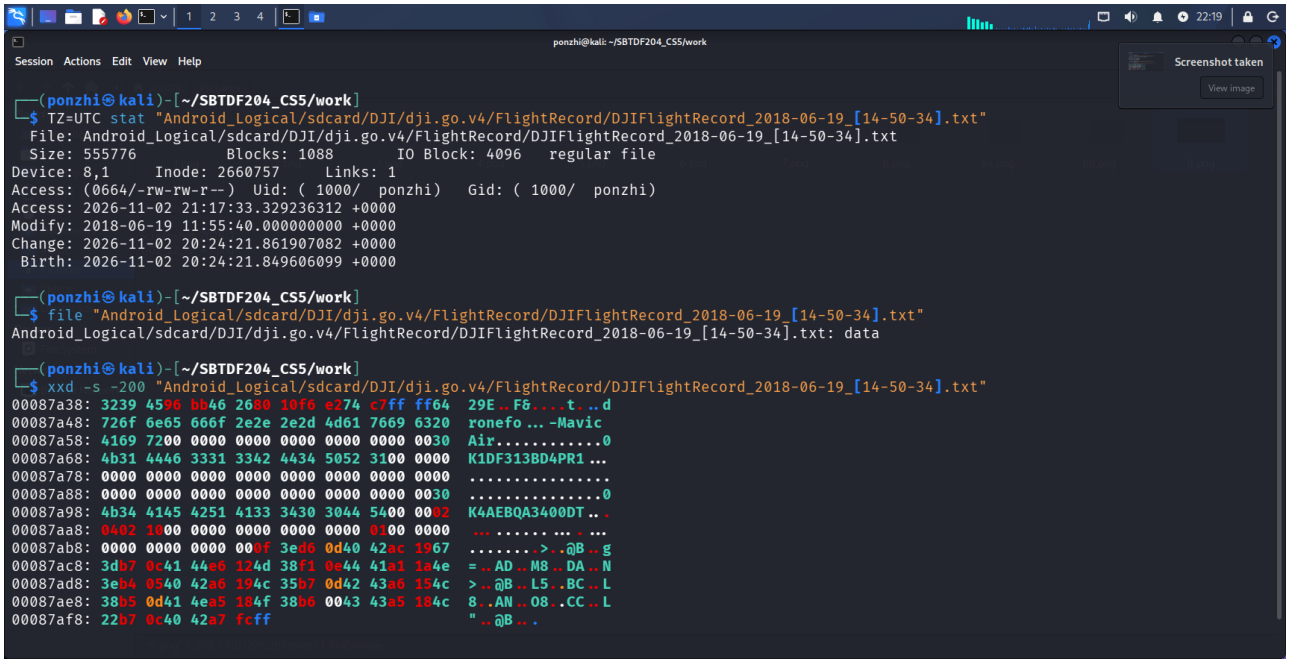
The file is named DJIFlightRecord and is 555,776 bytes in size. Its SHA-256 hash is 0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a. The data format is proprietary, so I cannot read the flight details directly. However, the last 200 bytes contain some readable text: "dronefo" and "Mavic Air," along with some non-printing characters and two short identifier strings, K1DF313BD4PR1 and K4AEBQA3400DT. This suggests that the aircraft is a Mavic Air, but I did not verify the model from any other fields.

The record ends at 11:55:40 UTC on 2018-06-19. This time marks the end of the flight record, from when the motors were turned on to when it landed.

The file name indicates a start time of 14:50:34, but this is based on the phone's clock. When comparing the timestamps, the phone clock is 2 hours and 55 minutes ahead of the file's timestamp. Therefore, I did not use the filename as proof of the start time.

To determine the flight duration, I need a parser. I've uploaded brief evidence to external sites, but I didn't extract data from the TXT without an approved offline parser. Based on my clock comparison, the phone clock runs 2 hours ahead of the file clock. Correcting for this means the end time of the TXT is 14:55:40 on the phone clock. The duration is 14:55:40 minus 14:50:34, which equals 5 minutes and 6 seconds. I consider this to be an inference.

The TXT file cannot be read for location information. The video info file has one GPS position: latitude 39.96119532709023, longitude -106.2164633258356, and altitude 12.0. This data supports the video, but does not provide a route from the TXT.



S10. TXT: stat, file and the last 200 bytes in hex. The access time now reads 2026-11-02 21:17:33 because of the earlier hashing.

Time-zone treatment and anomaly

My workstation is set to UTC+01:00. Three screenshots show this: the terminal clock shows 20:19, while the command "date -u" shows 19:19:23 UTC (S03, S04). The ls command prints +0100 and exiftool shows +01:00 (S12). The UTC values in my stat output come from using TZ=UTC stat, which adjusts the time to that zone.

The archive records local time. The course teaching deck shows the same file with a last write time of 12:55:40 -0400 on its machine, which converts to 16:55:40 UTC. The wall clock time is the same on both machines, but their offsets are different. This means the two UTC values are five hours apart. I believe the archive stores local time without a time zone, although I have not checked the archive. I will report two readings: Reading A is from my workstation (+01:00) and Reading B is from the US Eastern time zone (-04:00), as seen in the course deck. I do not state which is the correct UTC.

	Value as read	Clock	Source
TXT last write	2018-06-19 11:55:40	File time, reading A	S08, S10
MP4 last write	2018-06-19 11:55:42	File time, reading A	S08, S11
FLY057 last write	2018-06-19 11:58:24	File time, reading A	S08, S15
TXT file name	14-50-34	Phone clock	S07

	Value as read	Clock	Source
MP4 file name and CaptureDate	14:50:39	Phone clock	S07, S13a
Info file header line	Tue Jun 19 14:55:41 EDT 2018	Phone clock, label as written	S13
Workstation clock against date -u	20:19 against 19:19:23 UTC	Workstation, UTC+01:00	S03, S04

Question 4: What does the cached video establish?

The video file is located at Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4 and is 57,791,765 bytes in size. Its SHA-256 hash is 2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00. Next to it are two files: 2018_06_19_14_50_39.info and an analytics folder that contains.map and .tlv files. I noted these files but did not open them.

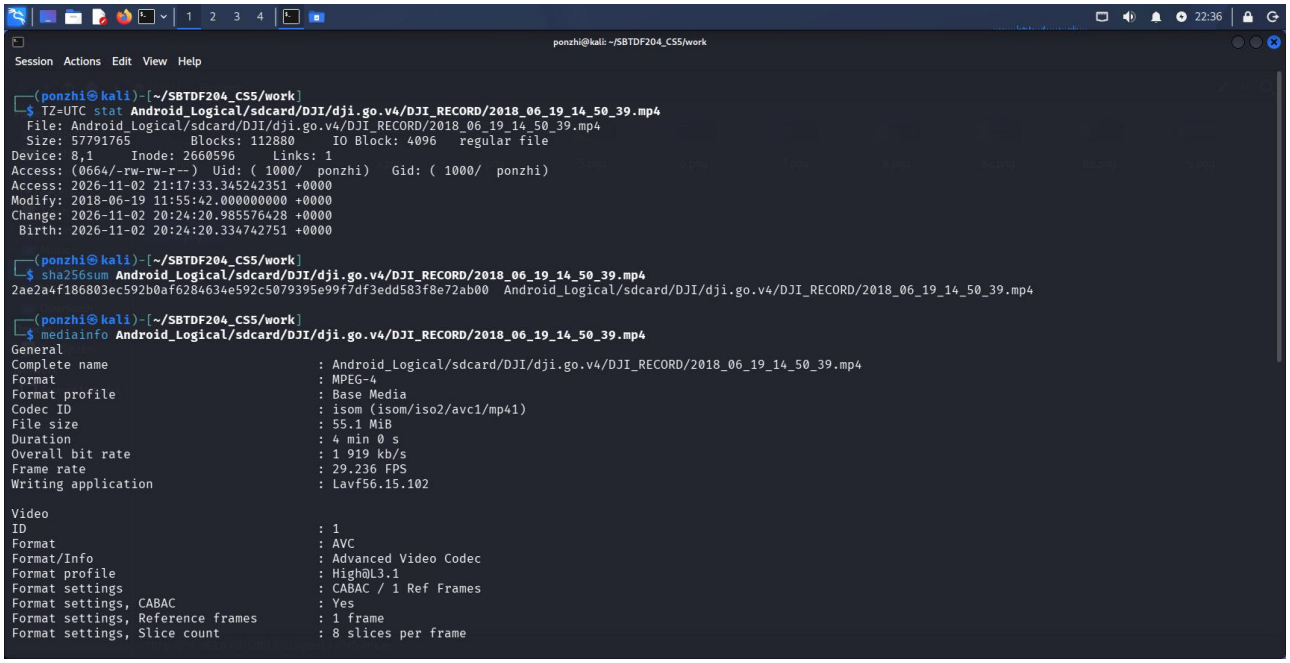
For the metadata, MediaInfo shows the video format is MPEG-4 with AVC High@L3.1 encoding. The resolution is 1280 by 720 pixels with a frame rate of 29.236 fps, and the duration is 4 minutes. The writing application used is Lavf56.15.102. Exiftool reports the duration as 0:04:01 and both the Create Date and Modify Date as 0000:00:00 00:00:00. The container does not have usable internal time, so the only times are from the file system and the info file. The info file states that the original drone video resolution is 1920 by 1080, while the local copy is 1280 by 720, meaning the file saved on the phone is the lower quality 720p copy.

The info file also lists the original phone path as /storage/emulated/0/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4, along with other details: CaptureDate=2018/06/19 14:50:39, EndTimeMsec=234833, PositionRelativeAlt=12.0, PositionGPSLat=39.96119532709023, PositionGPSLng=-106.2164633258356, PositionGPSAlt=12.0, and DeviceMaker=DJI.

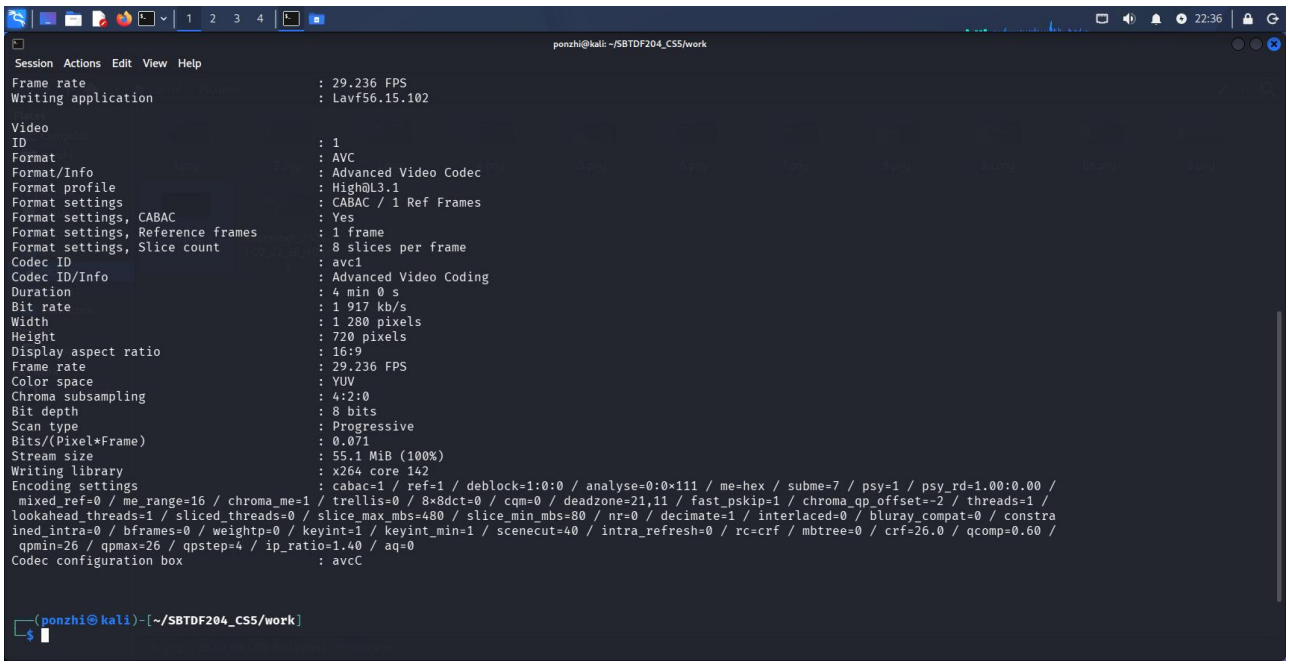
I recorded the file size and hash before opening the video. During playback, I used a copy named preview.mp4, which plays for 4 minutes and shows an outdoor scene with a white tent, vehicles, and several people.

In relation to the main flight record, the video was written 2 seconds after the TXT file (11:55:42 compared to 11:55:40). According to the phone clock, the video started 5 seconds after the TXT file (14:50:39 compared to 14:50:34). Both timings fit within the duration of the TXT.

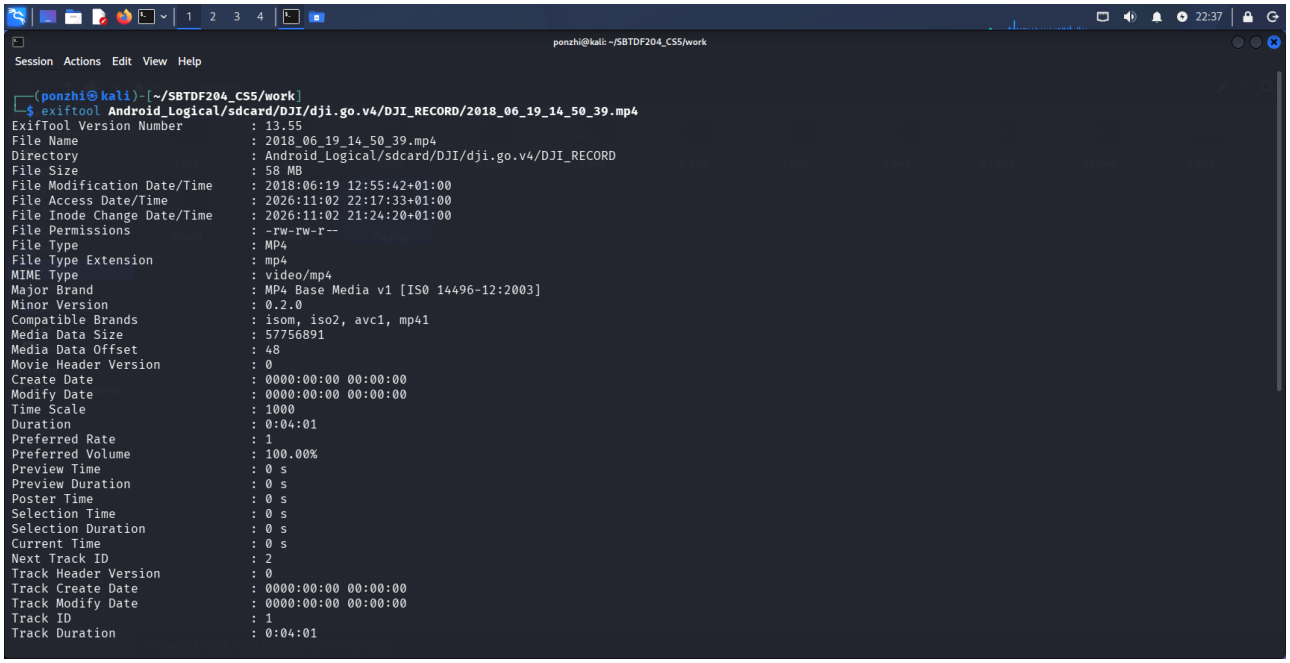
However, there is one point of confusion. The video duration is reported as 4 minutes by mediainfo, 4 minutes and 1 second by exiftool, and 3 minutes 54.8 seconds by EndTimeMsec. The phone clock shows a range from 14:50:39 to 14:55:41, which is 5 minutes and 2 seconds. The cached video file might not include the entire recording, but I have not confirmed that.



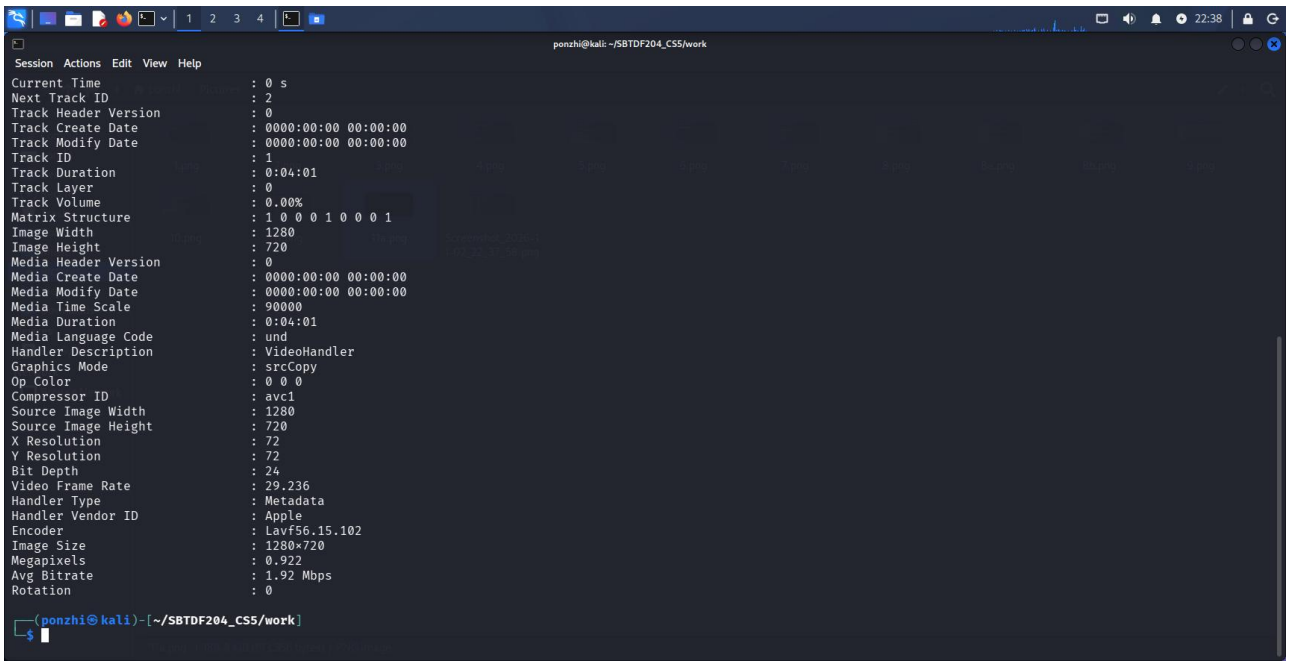
S11. MP4: TZ=UTC stat, sha256sum and the start of the mediainfo output.



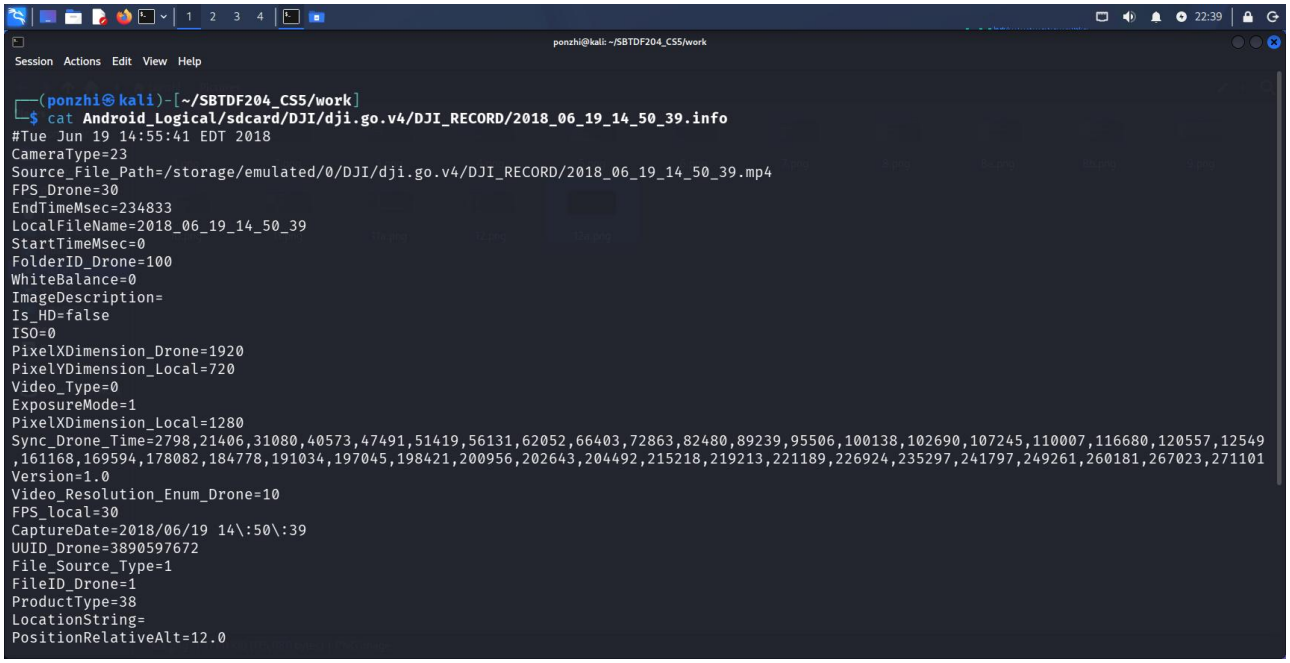
S11a. MP4: rest of the MediaInfo output; duration 4 min 0 s, 1280 by 720.



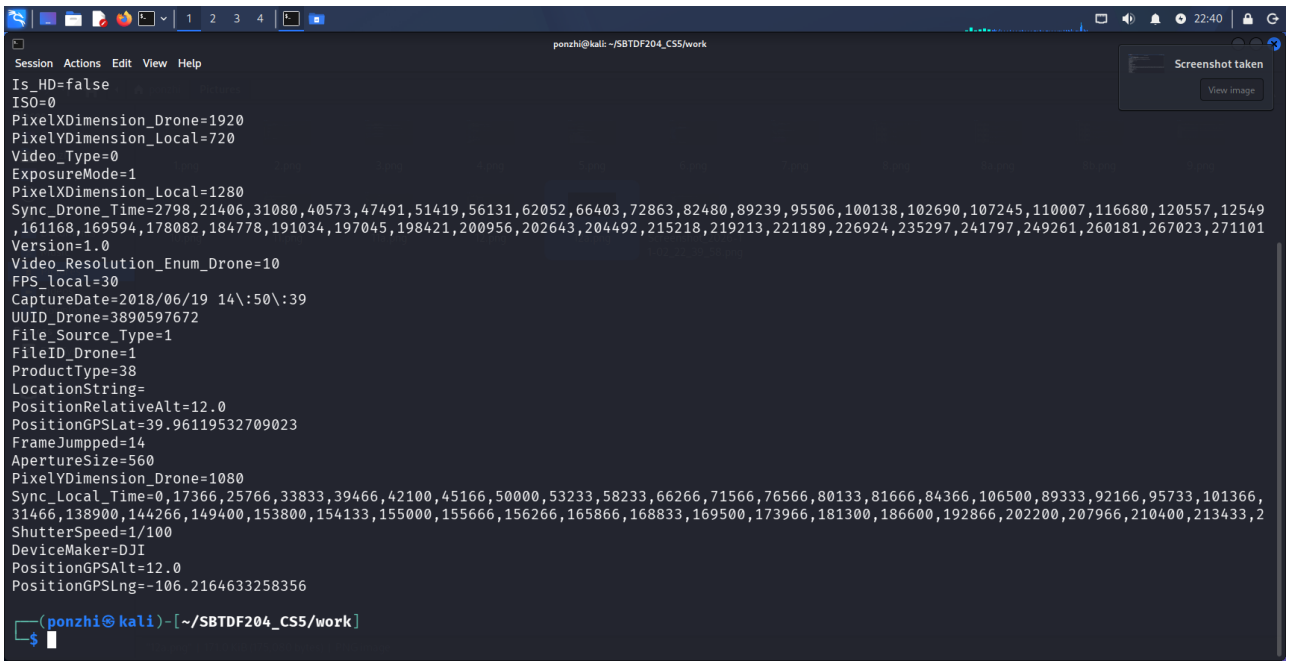
S12. MP4: exiftool output, part 1; modification time 2018:06:19 12:55:42+01:00, duration 0:04:01.



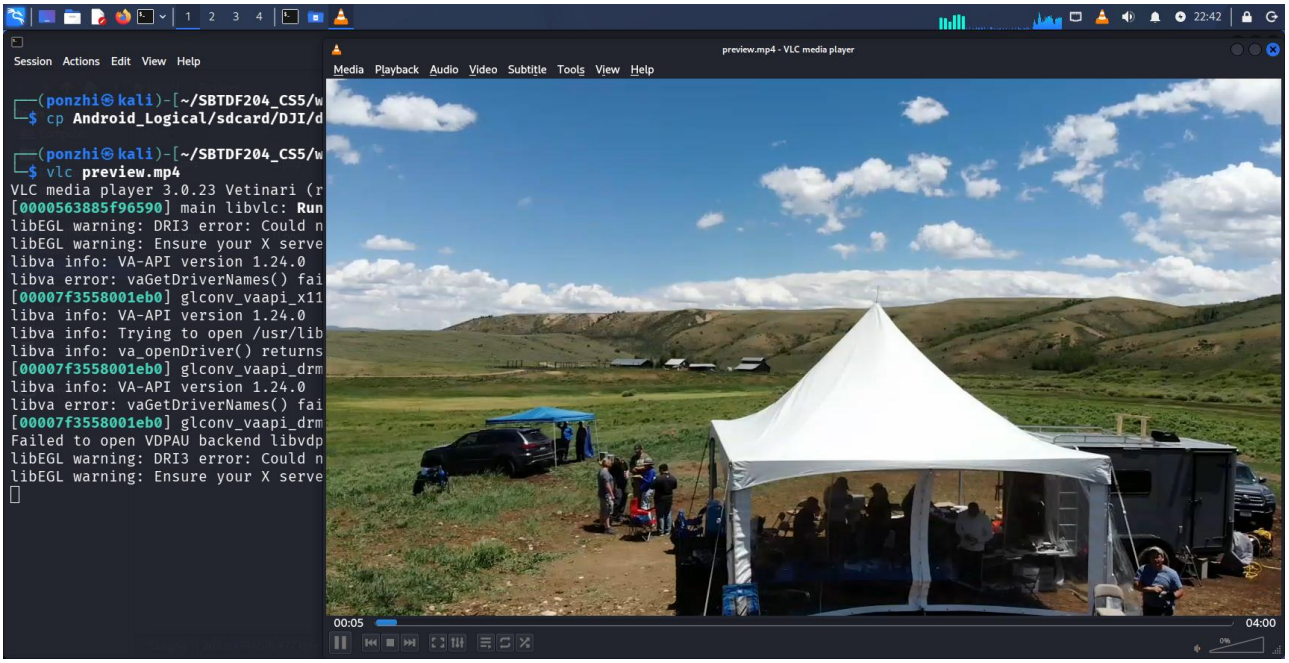
S12a. MP4: exiftool output, part 2.



S13. Video info file, part 1: header line, source path, EndTimeMsec, CaptureDate.



S13a. Video info file, part 2: GPS latitude, longitude and altitude.

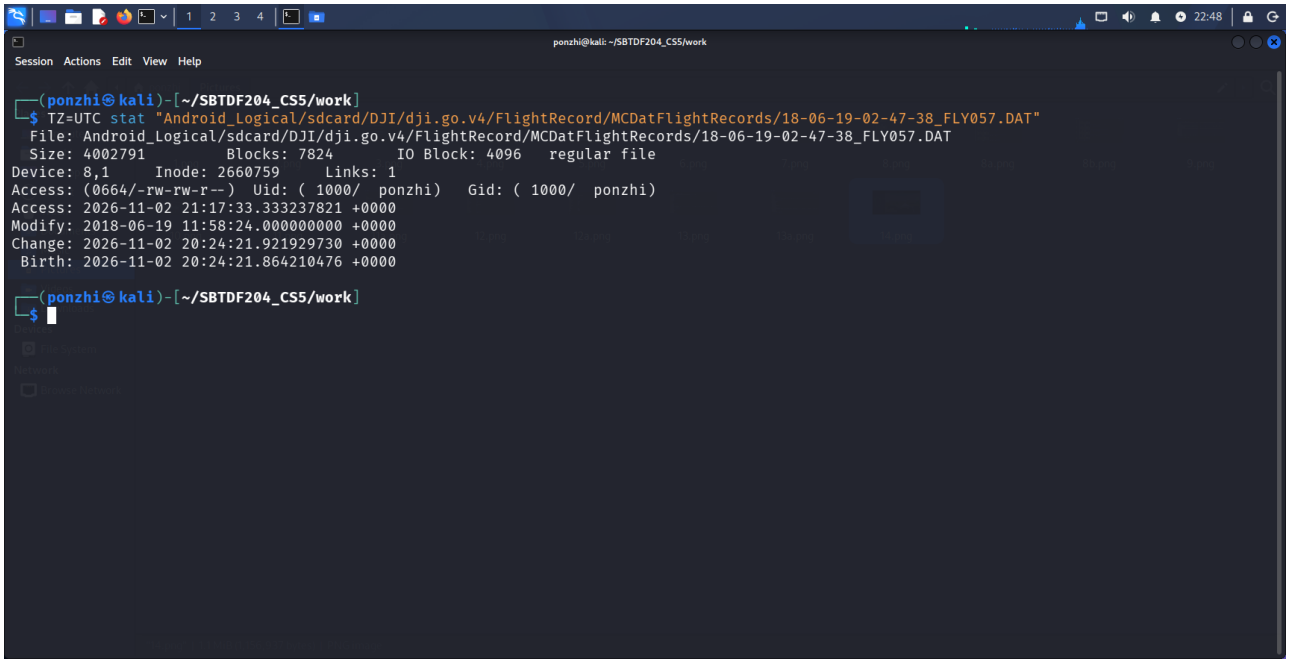


S14. Controlled preview of the copy preview.mp4 in VLC, total length 04:00.

Question 5: What does the detailed flight record establish?

There are two DAT files: 18-06-19-02-47-38_FLY057.DAT (4,002,791 bytes) and 18-06-19-02-59-05_FLY058.DAT (1,104,150 bytes). The DAT file records the time from when the device powers on to when it powers off. Therefore, it's normal to see a record that ends after landing, and that is what I found.

The hour in the DAT filenames (02) does not match any other time in the data. However, the minutes and seconds do. If we ignore the hour, FLY057 starts at xx:47:38 and ends at xx:58:24, covering a total of 10 minutes and 46 seconds. FLY058 starts at xx:59:05, which is 41 seconds later, and ends at 12:02:12, 3 minutes and 7 seconds after it started. Ignoring the hour, the first record starts about 12 seconds before the cached pictures were written (11:47:38 versus 11:47:50). These conclusions come from the file names and times. Without looking at the DAT content, I cannot confirm any events like power-on, motor-on, take-off, or landing. FLY058 seems like a second power cycle, but I have only looked at the times and hashes.



S15. FLY057.DAT: TZ=UTC stat.

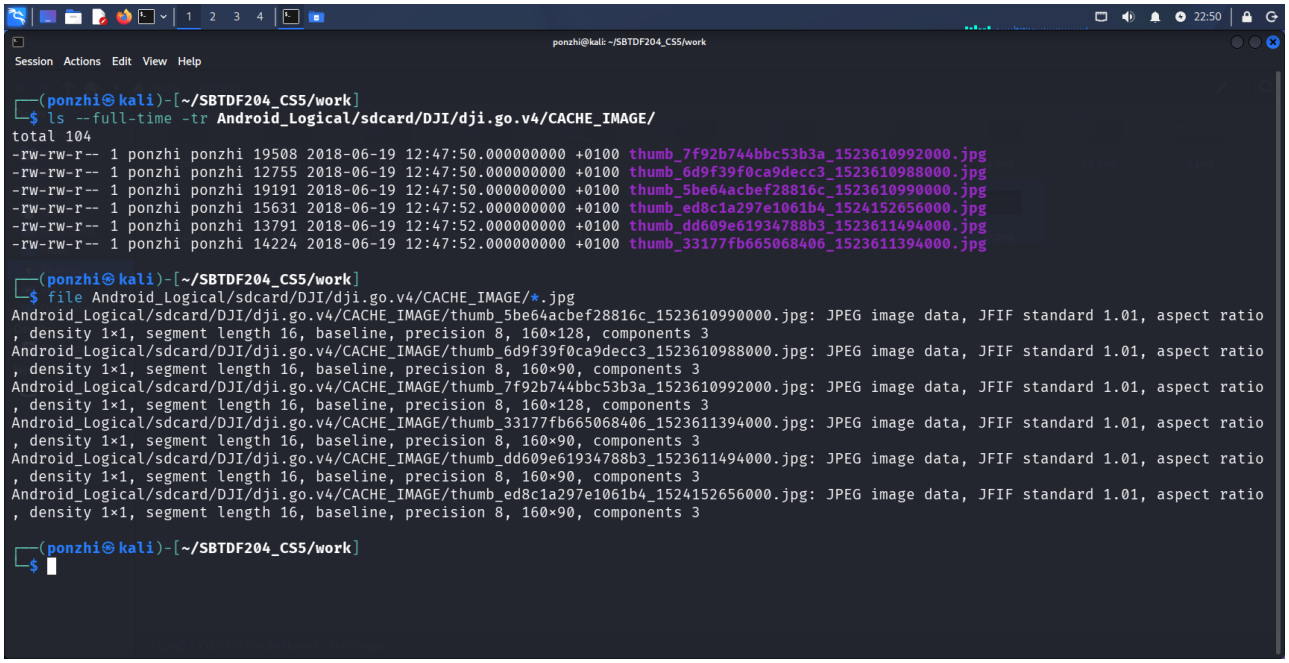
Question 6: What do the cached pictures establish?

CACHE_IMAGE contains six JPEG thumbnails (S07). The file shows baseline JFIF 1.01 images: two at 160 by 128 pixels and four at 160 by 90 pixels (S16). The file sizes range from 12,755 to 19,508 bytes.

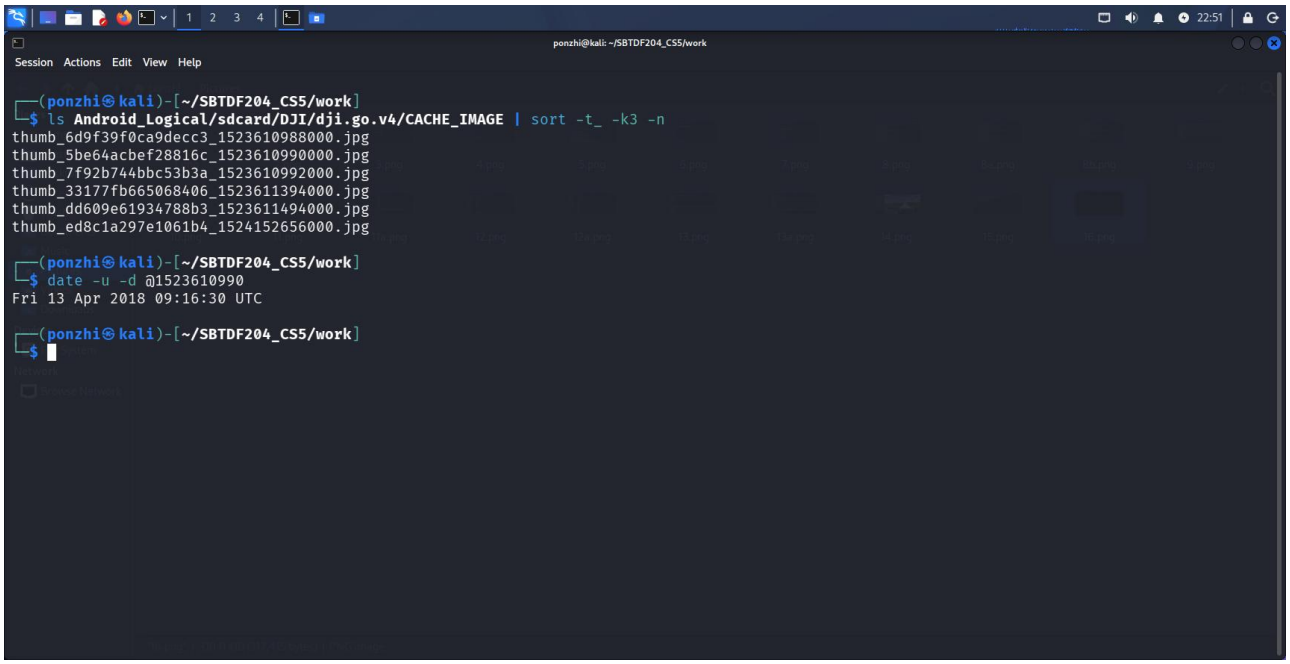
File times. The last write times for the files are 2018-06-19 at 11:47:50 for three files and 11:47:52 for the other three (S08a, S08b; S16 specifies these times as 12:47:50 and 12:47:52 +0100). The 13-digit number in each name represents Unix time in milliseconds. Using the command `date -u -d @1523610990`, we find it corresponds to Fri 13 Apr 2018 09:16:30 UTC (S17). My calculations show the other names correspond to 09:16:28, 09:16:32, 09:23:14, and 09:24:54 on 13 April 2018, and 15:44:16 on 19 April 2018.

Order. Based on the number in the name, the order is 6d9f, 5be6, 7f92, 3317, dd60, and ed8c (S17). By last write time (S16), the order is 7f92, 6d9f, 5be6, followed by ed8c, dd60, and 3317, with ties within the same second.

Limits. The names indicate April 2018, while the file times show 19 June 2018. One interpretation is that the name reflects the capture time, and the file time indicates when the app saved its cache, about 12 to 14 seconds after the power-on. Th app cache show what the app stored. They do not prove a flight, a location or an operator.



S16. CACHE_IMAGE: ls --full-time -tr and file output for the six thumbnails.



S17. CACHE_IMAGE: names sorted by the embedded number, and the decode of 1523610990.

Question 9: What conclusion can you defend?

- The Android image holds a TXT, an MP4 with an info file, two DAT files and six JPEG thumbnails under dji.go.v4.
- The TXT was last written at 11:55:40, the MP4 at 11:55:42 and FLY057 at 11:58:24. The same wall clocks read five hours.
- The file names and the info file are on a phone clock that differs from the file times by two hours (within one second).
- The cached video plays for 04:00 and shows an outdoor scene with several people. The info file holds one GPS position.
- The picture names decode to April 2018 and the picture file times are 19 June 2018.
- The TXT, MP4 and FLY057 belong to one flight session on 19 June 2018.

Timeline and correlation

Stage	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Device state and power on	E14	FLY057 name and last write	Name starts 02-47-38; last write 11:58:24	Hour in the name is unusable. Minutes and seconds only. Not confirmed without the DAT content.
Cached pictures written	E07, E15	Six thumbnails in CACHE_IMAGE	Three at 11:47:50, three at 11:47:52	App cache write time. Names decode to April 2018, so this is not a capture time.
Motors on, TXT start	E07, E09	TXT file name	Name reads 14-50-34 on the phone clock	Phone clock minus 2 h. Not proof of the start.
Take off or video start	E10, E12	MP4 name, CaptureDate	Name and CaptureDate read 14:50:39	Same correction. 5 s after the TXT name.
Main flight record event, landing	E07, E09	TXT last write	Modify 11:55:40	End of the motors-on to landed record.
Landing or video end	E07, E10	MP4 last write	Modify 11:55:42; player length 4 min 0 s	2 s after the TXT. Player length does not match the 5 min 2 s span.
Power off and correlation	E07, E14	FLY057 last write	Modify 11:58:24	2 min 44 s after the TXT. Fits power off after landing.
Second record end	E07	FLY058 last write	Modify 12:02:12	Not analysed beyond time and hash.

Corroboration. The three file types agree on order: video and TXT end together at landing, and the detailed record ends later at power off. Their names start within 5 seconds of each other on one phone clock. The inferred power-on sits 12 seconds before the cached pictures were written.

Conflict. The video length (4 min 0 s, 4 min 1 s, 3 min 54.8 s) differs from the 5 min 2 s span between its name and its info header. The picture names point to April 2018, while their file times point to 19 June. The two readings of UTC differ by five hours. The hour in the DAT names matches no clock.

Conclusion

The Android image suggests the following:

On June 19, 2018, DJI GO 4 on this phone stored records for one flight session. The video and main record ended at the same time, while the detailed record ended a few minutes later. These records include an app name and a device name but do not reveal who was flying the aircraft.

In terms of ownership, the data shows that the app was installed and used on this phone with a Mavic Air. However, there is no link between the phone, the aircraft, and any person. I did not check the property files.

Regarding operator identity, the cached video shows several people outside. Just appearing in the video does not indicate who flew the aircraft, and I did not attempt to identify anyone.

About the route, I could not read the TXT or DAT content. The only positional data I have is one GPS fix from the video info file, so I lack a route and distance flown.

The archive provides a wall-clock time. The UTC time depends on the zone used at extraction (there is a five-hour difference in the two readings), and the phone clock shows a time two hours ahead of the file clock. The absolute UTC time remains unclear.

Concerning cache limits, the cached video and thumbnails might be partial, re-encoded, or recorded at a different time than they were captured. The video lengths and picture names do not match their file times.

For handling effects, unpacking changed the creation and modification times, and my hashing changed the access times on the working copy. The first stat run (S08) shows the unmodified access times.

As for sensitive material, the video shows people, and the info file contains GPS coordinates. I only kept what was needed from the findings and did not publish the footage or search for the location on any external service.

Evidence log

Evidence ID	Tool and method	Source record, path or image	Observed fact
E01	command -v loop over 13 tools	Workstation	First run: mediainfo, vlc, ewfinfo, ewfverify, ewfmount missing. Second run: all present.
E02	sha256sum, sha256sum -c, chmod 444	originals/Android_Logical.zip	SHA-256 9a352b28...29285d; check OK; read-only.
E03	Version commands	evidence_out/tool_versions.txt	19:19:23 UTC 2 Nov 2026; sha256sum 9.10; ewfinfo 20140816; MediaInfoLib v26.05.
E04	cp, sha256sum *, unzip	work/Android_Logical.zip	Copy hash identical to E02; archive extracted.
E05	tree -L 1	Android_Logical/sdcard/DJI/dji.go. v4/	13 subfolders; tree prints 14 directories.
E06	tree (three runs)	FlightRecord, DJI_RECORD, CACHE_IMAGE	1 TXT, 2 DAT, 1 MP4 with .info, .map, .tlv, 6 JPEG.
E07	TZ=UTC stat on all artefacts	Ten artefacts (Appendix B)	Sizes, inodes, last-write times; access equals last write before hashing; birth and change 2026-11-02 20:24 UTC.

Evidence ID	Tool and method	Source record, path or image	Observed fact
E08	sha256sum on all artefacts	Ten artefacts	Ten SHA-256 values (Appendix B).
E09	stat, file, xxd -s -200	TXT	Last write 11:55:40; type "data"; text "dronefo" and "Mavic Air" in the last 200 bytes.
E10	stat, sha256sum, mediainfo	MP4	Last write 11:55:42; MPEG-4 AVC, 1280 by 720, 29.236 fps, 4 min 0 s.
E11	exiftool 13.55	MP4	Modification 2018:06:19 12:55:42+01:00; internal dates 0000:00:00; 0:04:01.
E12	cat	2018_06_19_14_50_39.info	Header 14:55:41 EDT; EndTimeMsec 234833; CaptureDate 14:50:39; GPS 39.96119532709023, -106.2164633258356.
E13	cp then vlc	preview.mp4 (copy)	Plays 04:00; outdoor scene with tent, vehicles, several people.
E14	TZ=UTC stat	FLY057 (and FLY058 in E07)	FLY057 last write 11:58:24, 4,002,791 bytes.
E15	ls --full-time -tr, file	CACHE_IMAGE	Six baseline JFIF JPEGs, 160 by 128 or 160 by 90; last write 12:47:50 to 12:47:52 +0100.

Mobile artefact hashes and times

File	Bytes	Inode	Last write	SHA-256
FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt	555776	2660757	11:55:40	0a9f1816d52a98f840f6e67c08a14473ae5a0a2e75e9484d5de66368c4fd4c4a
FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT	4002791	2660759	11:58:24	06130820a4a813bf21f7286150bd3c83084105fecc899bb4d8534895fa2df3a1

File	Bytes	Inode	Last write	SHA-256
FlightRecord/MCDatFlightRecords/18-06-19-02-59-05_FLY058.DAT	1104150	2660760	12:02:12	fdfccf0c5b1d90d189c19032bb34e151914f42fba42c7994d0492cf327ade07a
DJI_RECORD/2018_06_19_14_50_39.mp4	57791765	2660596	11:55:42	2ae2a4f186803ec592b0af6284634e592c5079395e99f7df3edd583f8e72ab00
CACHE_IMAGE/thumb_5be64acbef28816c_1523610990000.jpg	19191	2660588	11:47:50	e6a060c7756c7e2fd99c850434b02c708a329ebf16a814e0fe0a77b15caaab24
CACHE_IMAGE/thumb_6d9f39f0ca9decc3_1523610988000.jpg	12755	2660589	11:47:50	cbe6209aa64bbb44340e4aed5d2e276fd1df75805f956307b24a83e79b1d6f90
CACHE_IMAGE/thumb_7f92b744bbc53b3a_1523610992000.jpg	19508	2660590	11:47:50	32ff09733d14f4ecdd59171811fd50111cc00752ebdb528f04b75b08cf220874
CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg	14224	2660587	11:47:52	e8d6ac3d4cda2bccff4dd3e64950e87fd68700cf80b37f2c45e9e0ac59a4d42a
CACHE_IMAGE/thumb_dd609e61934788b3_1523611494000.jpg	13791	2660591	11:47:52	8986d1b4a162ca4bbe0adbf12f01141b9d7673ba196436ee31e16fb6b540b2e2
CACHE_IMAGE/thumb_ed8c1a297e1061b4_1524152656000.jpg	15631	2660592	11:47:52	924f798a8f815a2a8c3862bf40481f34194ce30fbd637c939e4aa66fcb108fd3